



SRH suppressed P-G-I design for very long-wavelength infrared HgCdTe photodiodes

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Abstract: The very long wavelength infrared (VLWIR, $> 14 \mu\text{m}$) spectral band is an indispensable part of new-generation infrared remote sensing. Mercury cadmium telluride (HgCdTe or MCT) has shown excellent potential across the entire infrared band. However, the dark current, which is extremely sensitive to the technological level and small Cd composition, severely limits the performance of VLWIR HgCdTe photodiodes. In this study, cut-off wavelengths of up to $15 \mu\text{m}$ for HgCdTe devices with novel P-G-I (including wide bandgap p-type cap layer, grading layer and intrinsic absorption layer) designs have been reported. Compared with a device with a double-layer heterojunction (DLHJ) structure, the designed P-G-I structure successfully reduced dark current by suppressing the Shockley–Read–Hall process. Considering the balance of quantum efficiency and dark current, with the introduction of an approximately $0.8 \mu\text{m}$ thickness Cd composition grading layer, the device can achieve a high detectivity of up to $2.5 \times 10^{11} \text{ cm Hz}^{1/2} \text{ W}^{-1}$. Experiments show that the P-G-I-T device has a lower dark current and a better SRH process suppressing ability than DLHJ devices, the measured detectivity achieved $8.7 \times 10^{10} \text{ cm Hz}^{1/2} \text{ W}^{-1}$. According to additional research, the trap-assisted tunneling current is the primary component of the dark current. Controlling the trap concentration to as low as $1 \times 10^{13} \text{ cm}^{-3}$ will be continuous and meaningful work. The proposed study provides guidance for VLWIR HgCdTe photodetectors.

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1. Introduction

According to Planck's radiation law, low-temperature objects are more likely to escape the tracing due to their low radiant exitance and long maximum exitance contrast wavelength. Therefore, very long wavelength infrared (VLWIR) photodetectors are essential for deep space exploration, exo-atmospheric target tracing, and special infrared (IR) spectral analysis units [1–3]. Owing to their low photon energy and easily activated band, only photodiodes with high quantum efficiency (QE) and low dark current can meet the requirements of high-precision and exorbitant equipment. Among the few extremely narrow bandgap materials, mercury cadmium telluride ($\text{Hg}_{1-x}\text{Cd}_x\text{Te}$ or MCT, x means Cd composition) has inherent advantages compared to band engineering quantum wells and superlattice IR detection materials. Because of its high absorption coefficient, tunable detection wavelength, and wide operating temperature range, HgCdTe has been widely used and researched [4–12]. Despite these advantages, the dark current behavior of HgCdTe photodiodes severely limits the performance of the detector owing to the small bandgap and immature technological level, particularly in the VLWIR range [13]. Several structural designs and process

optimizations have been proposed to improve the performance of LWIR and VLWIR HgCdTe photodetectors. Around the year 1990, the main purpose for researchers and engineers was to find appropriate the dopant and the material growth methods for LWIR HgCdTe photodiodes [14,15]. Peterman *et. al.* fabricated P-on-n (capital letter P means wide bandgap material) $\text{Hg}_{0.83}\text{Cd}_{0.17}\text{Te}$ electronic devices using molecular-beam epitaxy (MBE) technology, however, the fabrication of VLWIR photodiodes has not been applied [16]. With the advancement of material growth technology, MWIR and LWIR HgCdTe photodiodes can be fabricated using the liquid phase epitaxy (LPE) and MBE methods, leading to the development of second-generation infrared detection systems. As the process progressed, VLWIR HgCdTe photodetectors were fabricated and researched. Angelo *et. al.* in Raytheon Vision Systems reported in 2006 that they fabricated VLWIR HgCdTe photodiodes, which can respond to IR radiation up to 20 μm at 28 K. This study may have represented the most advanced technical level at that time, however further analysis of the photodiodes, detailed material growth, device fabrication, and structure design information have not been researched [17]. In 2010, Bertrand *et. al.* also reported VLWIR HgCdTe photodiodes working at 50 K [18]. The results indicate that devices with a p-on-n double-layer heterojunction (DLHJ) structure can achieve a better performance thanks to a longer minority carrier lifetime in the n-type doped layer and lower surface leakage current [19,20]. However, the material trap-related large dark current and bandgap offset induced low QE remain significant challenges for fabricating VLWIR HgCdTe photodetectors.

In this study, we report a novel P-G-I design for VLWIR HgCdTe photodiodes to suppress the Shockley–Read–Hall (SRH) process in the space charge region. The SRH process is an important dark current source for narrowband materials, and it is closely related to the traps and band gap [21]. This process frequently acts as a dominant component for HgCdTe devices at low temperatures [22,23]. When the Cd mole fraction changes the grading layer inserted, the device can maintain a low dark current and a good sensitivity despite the barrier on the valence band in the depletion region. The fabricated P-G-I photodiodes show lower generation-recombination (G-R) currents than DLHJ devices and remain high detectivity with appropriate designs. By analyzing the temperature-dependent dark current component of the two kinds of devices, we observed that the G-R current is suppressed and the trap-assisted tunneling current leads to a high dark current. Finally, further processing guidance for VLWIR HgCdTe photodiodes was proposed.

2. P-G-I design for VLWIR HgCdTe photodiodes

A VLWIR-sensitive material for HgCdTe is difficult to grow and process for easily introduced traps and damage. In addition, easily excited carriers are another noise source for VLWIR photodetectors. The DLHJ was designed to decrease the surface leakage and the n-type absorber layer can improve the minority-carrier lifetime. VLWIR DLHJ HgCdTe photodiodes are shown in Fig. 1(a). In the cap layer, the Cd composition was greater than 0.367, and in the absorption layer, the Cd composition was approximately 0.205. To further suppress the carrier generation and trapping involved in the SRH process, we designed a P-G-I structure with a grading layer inserted between the cap and the absorption layers, as shown in Fig. 1(b), where P represents the p-type wide-bandgap material, G represents the grading layer, and I represents the low-doping absorption layer. The key parameters for the structures are listed in Table 1. The main difference between the two structures is the band diagram near the junction, as shown in Fig. 1(c) and (d).

To investigate the relative merits of DLHJ and P-G-I designs for VLWIR HgCdTe photodetectors, finite element (FET) models were fabricated using Sentaurus TCAD. Figure 2(a) shows the simulated band diagrams of the HgCdTe photodiode with the grading layer thicknesses of 0 μm (DLHJ structure), 1 μm , and 2 μm , respectively. When the photodiode has a DLHJ structure, the holes in the n-type absorption layer can easily diffuse to the p-type cap layer to participate in conducting; therefore, R_0A is only 0.3 $\Omega\cdot\text{cm}^2$, as shown in Fig. 2(b). On the contrary,

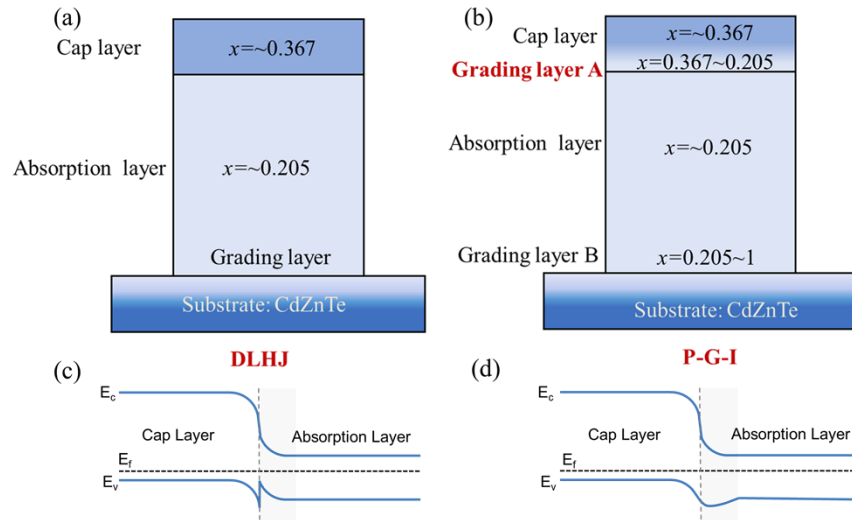


Fig. 1. Cross section of very long-wavelength DLHJ $\text{Hg}_{(1-x)}\text{Cd}_x\text{Te}$ photodiode (a) and P-G-I photodiode (b); Band diagram schematic of DLHJ photodiode (c) and P-G-I photodiode (d) at equilibrium state.

Table 1. Key parameters for the photodiodes with different structures

Structure	N_P (cm^{-3})	N_G (cm^{-3})	N_I (cm^{-3})	x_P	x_G	x_I
P-G-I	$>1 \times 10^{17}$	1×10^{15}	1×10^{15}	0.367	0.367–0.205	0.205
DLHJ	$>1 \times 10^{17}$	—	1×10^{15}	0.367	—	0.205

N means doping concentration and *x* means Cd composition

photogenerated carriers are easily transported across the junction and collected by contact. Assuming that the diffusion length in the absorption layer is greater than its thickness, it would have a high QE of more than 90%. When the thickness of the grading layer (Δx_G) was 2 μm , a large band offset was observed. In this situation, both minority carriers and photogenerated carriers can hardly transport the barrier under zero bias, so that the performance of large R_0A and low QE, will be observed. Between these two cases, the R_0A and QE will both have a discount. Nonetheless, this leads to an improvement in the detectivity, as shown in Fig. 2(c). The ideal specific detectivity was calculated by considering the Johnson noise and responsivity, and the peak specific detectivity was $2.5 \times 10^{11} \text{ cm} \cdot \text{Hz}^{1/2} \text{W}^{-1}$ with a grading layer thickness of approximately 0.8 μm . To reduce the effect of barriers on the transport of photogenerated carriers, an increase in the reverse bias voltage can flatten the band offset. Figure 2(d) shows the bias-dependent external quantum efficiency (EQE) for different grading-layer thicknesses (Δx_G), the light source for raytracing calculation is a Parallel light with a wavelength of 14 μm , the intensity of 1 W/cm^2 and the widow of $30 \times 30 \mu\text{m}^2$. The increase of EQE for lower bias is mainly because the barriers are flattening by the external bias. And after the transition voltage (the external bias when photocurrent approaches saturation), the barrier is nearly flattened and become invalid, therefore, the EQE is approaching saturation. Figure 2(e) shows the transition voltage of different photodiodes, it shows that a bias around -0.2 V can realize near-maximum QE of the photodiode with 0.8 μm grading layer, however for the photodiode with 2 μm grading layer, the transition voltage will exceed 1.6 V. Figure 2(f) shows specific detectivity when the diodes under -0.2 V bias, it can be observed that the 0.8 μm grading layer leads to the optimal sensitivity.

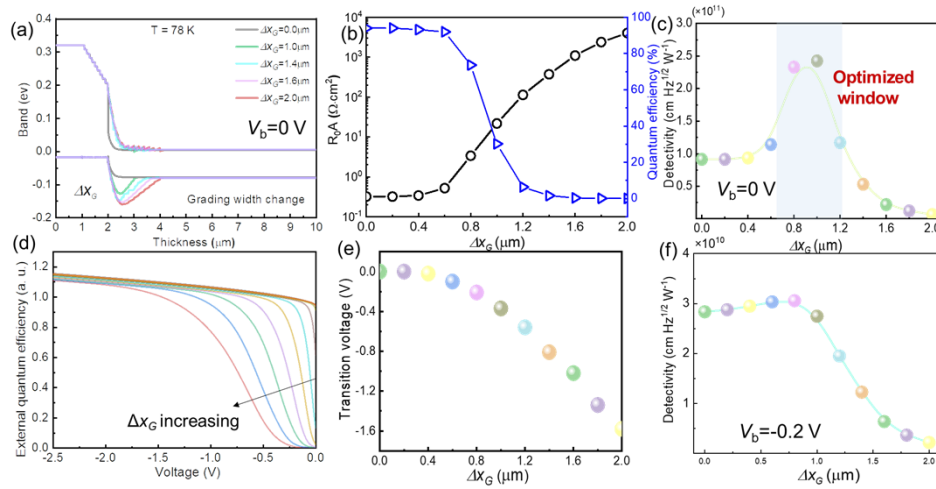


Fig. 2. Simulated band structure (a), R_0A and QE (b), and detectivity (c) at equilibrium state for different grading layer thicknesses; EQE (d) and transition voltage (e) of different photodiodes; (f) detectivities under -0.2 V .

Figure 3 shows the simulated dark current and SRH recombination rate distribution (balance state) for the VLWIR HgCdTe photodiodes with different grading layer thicknesses. For the DLHJ device, the large dark current is induced by a high SRH recombination rate in the depletion region. As the thickness of the grading layer increases, the SRH process in the depletion region is slightly reduced at the operating bias. Therefore, we can conclude that with appropriate grading layer thickness, and the P-G-I structure can improve the performance theoretically.

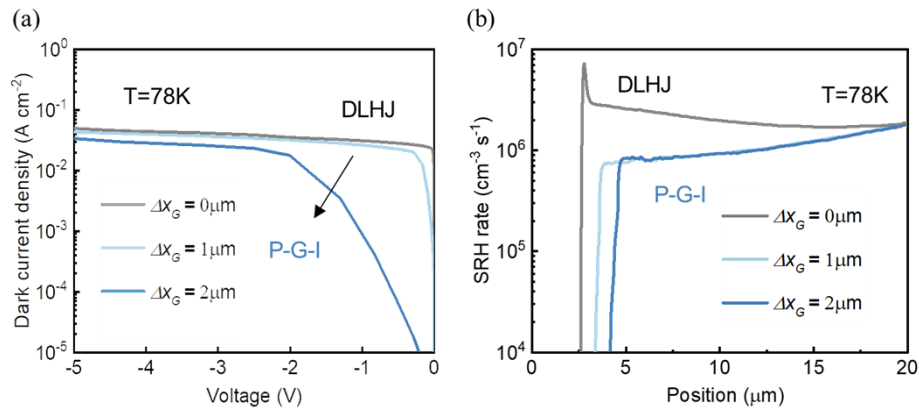


Fig. 3. Simulated dark current density (a) and SRH rate distribution (b) for the different structures.

3. Device fabrication and discussions

P-G-I VLWIR HgCdTe photodiodes were fabricated using LPE and conventional processes. Indium was used in the base layer as a donor by the telluride-rich LPE method, and arsenic was used in the cap layer as an acceptor by the mercury-rich LPE method. The two batches of photodiodes had the same size ($30 \times 30\text{ }\mu\text{m}^2$) and absorption layer thickness ($\sim 15\text{ }\mu\text{m}$). The p-area concentration is 10^{17} cm^{-3} orders of magnitude, and the n-area concentration is below $1 \times$

10^{15}cm^{-3} . In the absorption layer, the Cd composition was near 0.2, corresponding bandgap of approximately 83 meV at 78 K, and in the cap layer, the Cd composition was over 0.32. Limited by LPE technology, the thickness of the grading layer cannot be finely controlled. According to the measured results, the two batches of photodiodes can be named P-G-I-T (T indicates a thin grading layer) and P-G-I-W (W indicates a wide grading layer).

The I-V characteristic curves of two typical diodes from the two batches were measured with zero field of view (FOV) at different temperatures ranging from 50 K–140 K, as shown in Figs. 4(a) and (b). The laboratory-assembled setup consisted of a Keithley-263 voltage source, IEEE-488 bus, and HP4140B pA meter. Figure 4 shows that the P-G-I-T photodiode has better p-n junction rectification characteristics than P-G-I-W. In contrast to the P-G-I-W photodiode, it is obvious that the dark current characteristic of the P-G-I-T photodiode under a small reverse bias is temperature-dependent, which indicates that thermal-related diffusion and the G-R current are the main dark current sources of the P-G-I-T photodiode. Figure 4(c) shows the R_0A values at different temperatures of the two photodiodes, where the solid line is fit to the experimental data of P-G-I-T. The R_0A of the P-G-I-T photodiode at 78 K is $64\ \Omega\cdot\text{cm}^2$. Compared with the simulated R_0A , it can be observed that the grading layer is approximately 0.8–1.2 μm . From the fitted line of temperature-dependent R_0A , we extracted the activation energy (0.15 eV) of the P-G-I-T photodiode. This is larger than the bandgap of the absorption layer, which indicates that R_0A is dominated by the graded gap. The activation energy is larger than that of a device fabricated using DRS with a DLHJ structure (0.06 eV) [15]. In contrast, the P-G-I-W photodiode has a considerably larger R_0A of up to $264\ \Omega\cdot\text{cm}^2$ at 80 K, which indicates that the grading layer is larger than 1.2 μm . The temperature-dependent R_0A of P-G-I-W exhibits an increasing trend with large float, the exponential fitted activation energy is approximately 0.08 eV, which indicates that the energy is close to the absorbing material bandgap or the wideband material mismatch the depletion region. The IV characteristic of P-G-I-W has a resistive behavior rather than a photodiode one, this indicates that the dominant component of the reverse current is the larger tunneling current rather than the reverse saturation current. Figure 4(d) shows the measured blackbody responsivity of the P-G-I-T device with laboratory-assembled setup (as shown in the inset of Fig. 4(d)) consisting of blackbody source, SR570 pre-amplifier, and Signal Recovery 7270. The responsivity of P-G-I-T can achieve 9.67 A/W (500 K, 77 Hz) under zero bias and the corresponding QE is 67.7%. Figure 4(e) shows that the P-G-I-T devices have a cut-off wavelength of up to 15 μm at 78 K from the measured spectral response under $-50\ \text{mV}$ by VERTEX 80 FTIR Spectrometers. Importantly, the devices show the specific detectivities of $8.7\times 10^{10}\ \text{cm}\ \text{Hz}^{1/2}\ \text{W}^{-1}$ at 78 K background and $2.8\times 10^9\ \text{cm}\ \text{Hz}^{1/2}\ \text{W}^{-1}$ at 300 K background, respectively. However, the P-G-I-W device with high-quality parameters has no clear signal, which can be explained by the low QE as simulated.

We used an analytical method to further investigate the dark-current components of the two samples. The I-V curves were fitted by considering the thermal current (I_{th}), band-to-band tunneling current (I_{bbr}), trap-assisted tunneling current (I_{tat}) and surface leakage current (I_{surf}). The thermal current is essentially the combined contribution of diffusion and generation-recombination currents, which are of thermal origin. The following equation is a common description of the thermal currents in a P-n-heterojunction photodiode [24]:

$$I_{th} = I_{sat} \cdot \left(1 - \frac{V}{V_B}\right) \cdot \left[e^{\frac{qV}{\eta kT}} - 1\right] \quad (1)$$

where, I_{sat} represents the equivalent thermal saturation current of the junction diode which has a close relationship with the equivalent minority carrier lifetime in the depletion region, V_B means the potential difference caused by the band-offset, η represents ideality factor which represents the proportion of diffusion and G-R current. For narrow-bandgap materials, the tunneling current is triggered by a small outfield effect even at zero bias [25]. The direct tunneling current can be

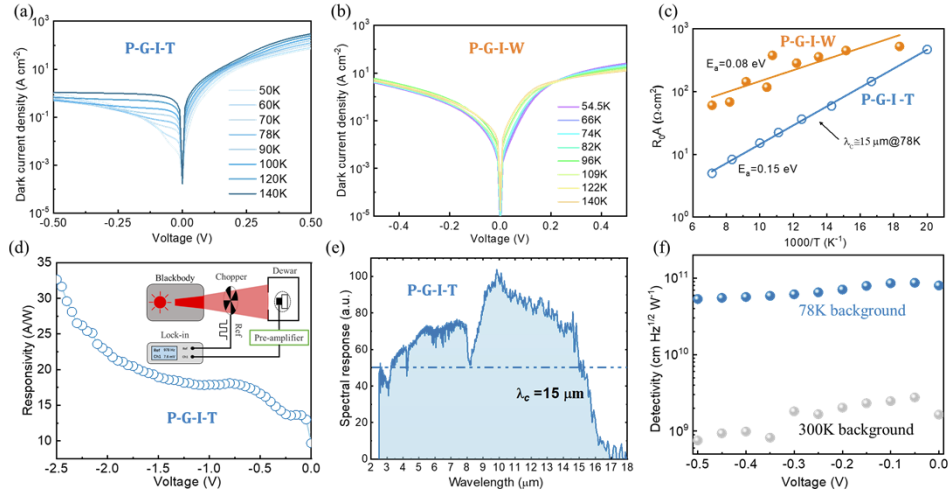


Fig. 4. Measured temperature-dependent properties IV of the two typical samples: (a) and (b) are the IV curves, (c) means the R_0A product parameter; (d) Blackbody responsivity, (e) normalized spectral response and (f) detectivity of the P-G-I-T sample.

expressed as follows [26,27]:

$$I_{bbt} = -\frac{Aq^3\sqrt{2m^*}\xi(V_{bi} - V)}{4\pi^3\hbar^2\sqrt{E_g}} \cdot e^{-\frac{\pi\sqrt{m^*}/2E_g^{1.5}}{2q\xi\hbar}} \quad (2)$$

the indirect, that is, trap assisted tunneling current is expressed as follows [21]:

$$I_{lat} = -\frac{A\pi^2q^2N_t m^* \xi^c M^2}{\hbar^3(E_g - E_t)} \cdot e^{-\frac{8\pi(m^*)^{0.5}(E_g - E_t)^{0.5}}{3q\xi^c\hbar}} \quad (3)$$

where A represents the area, m^* represents the effective mass, ξ represents the electric field strength, M represents the matrix element associated with the trap potential, N_t represents the trap density, E_t represents the trap level position, c represents a fitting parameter, and all other symbols have their usual meanings. The surface leakage mechanism plays an important role in dark current, especially for the III–V infrared materials. We regard the surface leakage current as shunt-would [28]. Given the afore mentioned dark current component, the total fitted dark current can be written as follows [29]:

$$I_{fit} = I_{th} + I_{bbt} + I_{lat} + I_{surf} \quad (4)$$

By fitting the measured dark currents, we obtained the detailed components of the two samples. When the diodes are operated at a large reverse bias, the space charge region will extend into the absorption region, while the two samples have the same bandgap and doping concentration, therefore, their band-to-band tunneling currents are equal. It is not suitable for the device working under this bias. Additionally, the trap-assisted tunneling current dominated the main IV characteristic difference between the two samples under a large reverse bias, as shown in Fig. 5(c). It indicates that the material quality of P-G-I-T is better than that of P-G-I-W. The key parameters extracted for the two types of diodes at 78 K are listed in Table 2. The comparison of these key parameters was under the same voltage, this would be useful for pursuing the improvement of one indicator (such as dark current), but not the signal to noise ratio which should be compared at the same QE. The reverse saturation current of the P-G-I devices was lower than that of the DLHI

device. Although the P-G-I-T and P-G-I-W devices have the same absorption layer, the larger barrier in the space charge region of P-G-I-W device blocks the diffusion of minority carriers, leading to the difference of thermal saturation current. The ideality factor of the P-G-I-T device was close to 1, indicating that the G-R process was suppressed. The larger ideality factor, trap concentration, deep trap level, and the decreasing dark current under the middle reverse bias region of P-G-I-W indicate that the trap-related tunneling process is dominant, even under zero bias. The surface current in our calculation is comparable small (equivalent parallel resistance is over 300 k Ω), this indicated the structure still has the ability of suppressing the surface leakage mechanism.

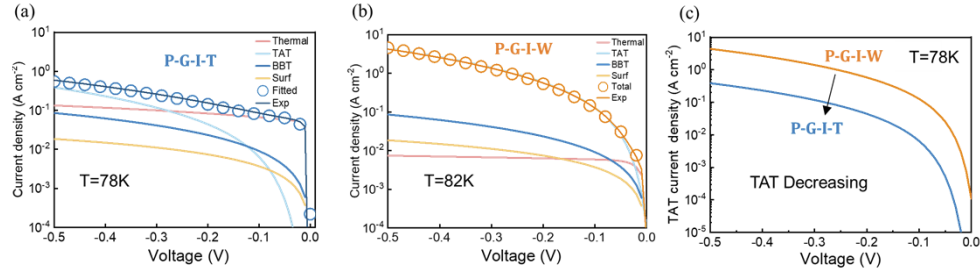


Fig. 5. Dark current analysis results obtained by the proposed analytical method: (a) and (b) means the dark current component; (c) comparison of extracted TAT current density of the two samples.

Table 2. Extracted parameters for the photodiodes with different structures

Structure	I_{sat} (A)	η	N_t (cm $^{-3}$)	E_t (eV)
P-G-I-T	4.5×10^{-7}	1.2	$\sim 3 \times 10^{15}$	0.04
P-G-I-W	0.5×10^{-7}	2.6	$\sim 4.6 \times 10^{16}$	0.06
DLHJ (Ref. [15])	$\sim 5 \times 10^{-6}$	1–2	$< 3 \times 10^{15}$	0.06

To quantitatively analyze the influence of traps, we simulated the dark current of the photodiodes with different trap concentrations using TCAD, as shown in Fig. 6(a) [30,31]. With an increase in trap concentration, there is a larger capture cross-sectional area involved in the tunneling and recombination processes. Therefore, the dark current was consistent with the trap concentration. In addition, detectivity of the photodiodes under -0.5 V bias is calculated as shown in Fig. 6(b), which exhibits a monotonically decreasing property with the trap concentration. This indicates that controlling traps is still an important task.

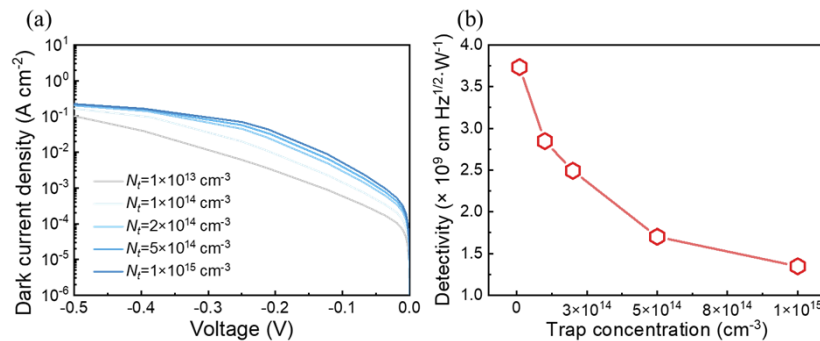


Fig. 6. (a) Dark current of the photodiodes with different trap concentrations; (b) detectivity of the photodiodes under -0.5 V bias with different trap concentrations.

4. Conclusions

In summary, a P-G-I design for VLWIR HgCdTe photodiodes was proposed and optimized to suppress SRH process. The simulated results show that with a near $0.8 \mu\text{m}$ grading layer inserted, the device exhibits better performance than the DLHJ HgCdTe photodiodes. The fabricated P-G-I-T device with an opportune grading layer had a cut-off wavelength of up to $15 \mu\text{m}$ at 78 K , and the detectivity is as high as $8.7 \times 10^{10} \text{ cm Hz}^{1/2} \text{ W}^{-1}$. The P-G-I-T device also showed a higher activation energy, leading to successful suppression of the G-R current. From the measured and fitted I-V curves, the key parameters are extracted, and the saturation current, ideality factor, and trap level position for the P-G-I-T device are better than those of the DLHJ device in reference. In addition, we observed that the TAT current is still one of dominant dark current components, and controlling the trap concentration as low as $1 \times 10^{13} \text{ cm}^{-3}$ could greatly improve the performance of VLWIR HgCdTe devices. This study may extend the applications of VLWIR photodetectors by lowering background doping and new concept bandgap designs.

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Data availability. The data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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